



Eco-Visual Branding for Lake Perintis Through Plastic Waste Transformation

**Syarifah Fatimah Setiasih Niode^{1*}, Wira Pratama Rumambie², Muhammad Akram Mursalim³,
Moh. Imron Rosidi⁴, Abd Halid Lemba⁵**

¹⁻⁵ Universitas Negeri Gorontalo, Indonesia

email: syarifahfnsiode@ung.ac.id¹

Article Info :

Received:
20-08-2026
Revised:
15-09-2026
Accepted:
02-10-2026

Abstract

Danau Perintis faces interconnected challenges involving plastic waste management and the limited availability of distinctive tourism merchandise that represents the destination. This study develops an eco-visual branding approach by transforming discarded plastic bottle caps into functional keychains that combine local visual identity, ecological narratives, and practical utility. An exploratory research and development design was implemented through the Double Diamond stages of Discover, Define, Develop, and Deliver, supported by field observation, informal manager interviews, material experimentation, prototyping, documentation, simulated market testing, and descriptive evaluation. The development produced twelve keychains under the brand identity Perintis by Aruna with the tagline "From Waste to Worth." All twelve units were sold at IDR 20,000 per unit during a campus-based exhibition, generating IDR 240,000 in gross revenue and providing initial evidence of product acceptance within the tested environment. The findings further demonstrate that material transformation can function as part of destination storytelling by connecting waste origin, local visual elements, product utility, and consumer interaction. The study formulates the Eco-Visual Branding Loop, integrating ecological input, material transformation, local visual identity, digital storytelling, AISAS activation, and market feedback. The model provides a practical framework for developing circular tourism merchandise while requiring further validation through larger-scale production, consumer profiling, material safety testing, and longitudinal digital communication measurement.

Keywords: Circular Tourism, Digital Storytelling, Eco-Visual Branding, Tourism Merchandise, Waste Transformation.



©2026 Authors.. This work is licensed under a Creative Commons Attribution-Non Commercial 4.0 International License.
(<https://creativecommons.org/licenses/by-nc/4.0/>)

INTRODUCTION

Tourism destinations are perceived not only through physical landscapes and services but also through sign systems that construct visitor memories, emotional associations, and interpretations of place. Visual identity, local storytelling, merchandise or souvenir products, and digital content operate synergistically to guide how tourists recognise, interpret, and internalise a destination, making visual communication increasingly inseparable from the construction of destination meaning. Destination brand experiences established across official online platforms can influence perceived authenticity and subsequent behavioural responses because digital representations shape destination expectations before, during, and after the visit (Jiménez-Barreto et al., 2020). Recent research further conceptualises destination branding as a co-created gestalt in which multiple visual, experiential, social, and symbolic elements are assembled into a holistic perception of place rather than communicated through an isolated brand artefact (Wang et al., 2025). Visual branding therefore functions not merely as an aesthetic mechanism but as an interface through which landscapes, local meanings, visitor experiences, and market interactions become connected. Evidence from Indonesian tourism destinations also indicates that visual branding can influence tourists' perceptions, although its effectiveness depends on how visual elements are connected to destination characteristics and visitor interpretation (Prawira et al., 2023). The contemporary challenge is consequently to develop destination identities that remain visually recognisable while simultaneously communicating authenticity, sustainability, and a meaningful relationship between visitors and the places they encounter.

Danau Perintis, located in Bone Bolango Regency, Gorontalo Province, possesses an attractive natural landscape and site planning that has attracted domestic visitors, yet its existing tourism

experience has not fully translated environmental resources into distinctive brand assets. Initial field observations revealed two structural challenges: inorganic waste, particularly plastic bottles and bottle caps, is predominantly collected and incinerated, while the destination lacks distinctive merchandise that connects visitor experience with local identity. This condition reflects a wider difficulty in Indonesian tourist attractions, where waste recycling remains constrained by operational, infrastructural, behavioural, and policy-related challenges despite restrictions on single-use plastics and increasing environmental awareness (Suryawan et al., 2024). Recent evidence from an Indonesian freshwater tourism destination also demonstrates that lake debris is connected with tourist acceptance and sustainability management, indicating that waste should be treated as part of destination governance and visitor experience rather than as an isolated sanitation problem (Iswahyudi et al., 2026). At the same time, community-based approaches to plastic waste management emphasise the importance of integrating local actors, material recovery, and tourism development so that environmental interventions generate value beyond waste reduction itself (Nizar et al., 2026). For Danau Perintis, the problem therefore concerns not simply how plastic is removed from the tourism environment, but how discarded material can be redirected into a productive system that creates environmental, cultural, and communicative value. Reframing plastic bottle caps as design resources opens an applied innovation pathway in which waste management can become directly connected to destination identity and visitor engagement.

Plastic waste management at tourism sites needs to be addressed within the circular economy framework because conventional collection and disposal retain the destination within a predominantly linear material system. The circular economy shifts the take-make-dispose paradigm towards maintaining material value through reuse, recycling, recovery, and design innovation, although tourism applications remain comparatively emergent and fragmented across different destination contexts (Rodríguez et al., 2020). Within tourism and hospitality, circular practices can generate environmental and business benefits, but their effectiveness depends on stakeholder participation and their integration into broader destination value chains rather than on isolated recycling activities (Jones & Wynn, 2022). Indonesian initiatives around Borobudur similarly illustrate how plastic upcycling can be embedded in tourism areas to transform waste management into participatory creative activity rather than treating discarded material exclusively as an environmental burden (Khaerunnisa & Cininta, 2023). The relevance of this approach to Danau Perintis lies in the possibility of converting discarded bottle caps into functional and symbolic products whose material history remains connected to the destination. Such transformation requires design decisions concerning form, usability, visual identity, narrative, and perceived value rather than simple material substitution. Design thinking provides a useful foundation for this process because it frames innovation as an iterative response to human needs, contextual problems, and opportunities for creating meaningful solutions (Brown, 2008).

Developing souvenirs from upcycled materials demonstrates strong potential to link environmental management, creativity, and regional identity because the resulting product can simultaneously function as a material recovery outcome and a communication medium. Research on Surakarta demonstrates that recycled plastic can be transformed into souvenir products associated with city-specific identity, showing how waste materials can acquire cultural and economic value through intentional design (Wahida & Fuad, 2022). A comparable study using waste-derived biocomposite materials positions sustainable tourism souvenirs as instruments of green innovation and place branding, indicating that the material composition of a product can become part of its destination narrative when design and branding are developed together (Syaputra et al., 2024). Yet these approaches leave an important conceptual space between the physical transformation of waste and the subsequent circulation of the product through visitors' digital experiences. Souvenirs are not simply commercial objects because their locality can be constructed through representations of place, people, and remembered experiences, enabling them to extend destination meaning beyond the temporal boundaries of the trip (Liu et al., 2025). Their branding potential is also strengthened when authenticity and visitor co-creation contribute to destination brand experience and souvenir repurchase intention (Jog & Gumparthi, 2025). The unresolved issue is how an upcycled souvenir can operate simultaneously as a circular-economy product, a visual representation of a specific destination, and a trigger for continued digital interaction after visitors leave the site.

The communicative dimension becomes particularly important because sustainability-oriented products do not automatically generate positive consumer responses merely by displaying

environmental claims. Sustainability communication in tourism has long been characterised by a tension between the increasing prominence of environmental discourse and the difficulty of translating sustainability messages into memorable and persuasive visitor experiences (Tölkes, 2018). Tourist responses can become more meaningful when sustainability is embedded within the experience rather than communicated as an overly explicit promotional message, suggesting that ecological value should be integrated into the product narrative and visual experience itself (Hanna et al., 2018). From this perspective, the narratives “Dari Sampah Jadi Kisah” (“From Waste to Story”) and “From Waste to Worth” position ecological transformation as an intrinsic component of product meaning rather than as a moralising claim. The cultural dimension also requires careful consideration because destination branding that extracts local symbols without adequate contextualisation may weaken authenticity and create tensions around cultural ownership and representation (Freire et al., 2022). This concern reinforces the need for an eco-visual identity in which local visual elements are selected and transformed through contextual design rather than being applied as superficial decoration. The product consequently needs to communicate three interconnected meanings: the environmental transformation of discarded plastic, the visual identity of Danau Perintis, and the visitor's continuing relationship with the destination.

On the promotional front, the management of Danau Perintis relies predominantly on Facebook, which provides a relevant communication channel for local audiences but does not yet constitute a structured content ecosystem connecting product identity, information search, purchase action, and post-purchase sharing. Social media marketing enables organisations to develop interactive relationships, electronic word-of-mouth, consumer engagement, and brand-related responses, making digital platforms important environments for extending destination communication beyond physical encounters (Alalwan et al., 2017). The AISAS model—Attention, Interest, Search, Action, and Share—is particularly relevant because it conceptualises digital consumer behaviour as a sequence in which audiences can move from exposure and interest towards information seeking, action, and subsequent sharing (Abdurrahim et al., 2019). Research on video branding in ecotourism demonstrates that AISAS can connect visual content with tourism communication by encouraging audiences to process, search, respond to, and disseminate destination information through digital media (Fadlurrahman et al., 2023). Similar evidence from TikTok shows the relevance of AISAS to contemporary short-form social media communication, where attention is rapidly converted into search, interaction, action, and sharing behaviours (Sasmita & Achmadi, 2022). These findings indicate that the communication value of an eco-souvenir should not terminate at the point of purchase because the product itself can become a visual trigger for digital storytelling, user-generated content, and post-visit destination remembrance. The research gap therefore concerns the absence of an integrated framework that connects plastic waste transformation, local visual branding, souvenir design, AISAS-based digital communication, and market feedback within one circular destination communication cycle. This study addresses that gap by developing the concept of eco-visual branding, positioning the upcycled product as a tangible message carrier that connects the physical destination, environmental action, local identity, digital storytelling, and post-visit engagement. The study aims to explain the development process of tourism merchandise crafted from plastic bottle caps, evaluate its initial market acceptance through production and sales data within a simulated market environment, and formulate the Eco-Visual Branding Loop as an initial framework for circular destination branding. Its theoretical contribution lies in extending destination branding and circular tourism perspectives by conceptualising waste-derived merchandise as a simultaneous material, visual, and communicative asset, while its methodological contribution lies in integrating design development, product-market feedback, and AISAS-oriented communication into an applied framework that can be examined and refined in subsequent destination studies.



Figure 1. Danau Perintis as the Locational Context for Eco-Visual Branding Development
Source: Personal Documentation (2025)

RESEARCH METHODS

This study employed an empirical exploratory Research and Development (R&D) design guided by the Double Diamond framework, which consists of the Discover, Define, Develop, and Deliver stages and provides a structured process for moving from problem identification toward iterative solution development (Brown, 2008). The study was conducted at Lake Perintis, Bone Bolango Regency, Gorontalo Province, with the development process integrated into Research-Based Learning involving undergraduate communication students in field observation, concept formulation, prototype production, promotion, simulated sales, and evaluation. The Discover stage involved direct observation of plastic bottle-cap waste, existing waste-handling practices, destination promotional media, visitor characteristics, and potential consumer preferences, supported by informal interviews with destination managers. Findings from this stage were translated during the Define stage into the positioning, visual identity, naming, tagline, colour palette, product form, and narrative concept of *Perintis by Aruna*. The Develop stage experimentally transformed collected High-Density Polyethylene (HDPE) and Polypropylene (PP) bottle caps through sorting, washing, drying, heat melting, moulding, trimming, hardware assembly, and packaging, following a circular approach that seeks to retain material value through reuse and recovery within tourism activities (Rodríguez et al., 2020). The Deliver stage consisted of prototype presentation, promotional documentation, simulated market testing, sales activities, and evaluation, enabling the developed merchandise to be examined as both an upcycled material product and a destination-branding medium.

Table 1. Stage of Eco-Visual Branding Development

Stage	Main Activities	Output	Key Indicators
Discover	Waste observation, manager interviews, target-audience analysis	Problem and needs map	Identified waste material, promotional channels, and user profiles
Define	Positioning, visual identity, and messaging formulation	<i>Perintis by Aruna</i> concept	Formulated name, tagline, form factors, and colour palette
Develop	Material experimentation, prototyping, and packaging design	Keychain prototypes	Functional products carrying local narratives
Deliver	Simulated market testing, forum presentation, documentation, and evaluation	Initial market data and conceptual loop	Sales response, consumer feedback, and prototype evaluation

Source: Research Data (2025)

Validation combined field-observation records, manager information, visual documentation, production logs, sales records, and informal consumer feedback to establish evidence triangulation across the development process. Prototype quality was evaluated according to identity legibility, aesthetic appeal, functional usability, durability, narrative alignment, material transformation, and production replicability, recognising that the emotional and experiential qualities of designed objects contribute to how users perceive and value products (Norman, 2004). The environmental dimension was assessed by examining the extent to which discarded bottle caps could be transformed into functional merchandise while minimising secondary production residues and maintaining a coherent sustainability narrative, reflecting the importance of communicating tourism sustainability through meaningful experiences rather than through environmental claims alone (Tölkes, 2018). Initial market acceptance was evaluated descriptively using production quantity, available inventory, number of units sold, sales completion, observed purchasing interactions, and informal consumer responses concerning product form, visual appearance, usefulness, price, and destination association. The AISAS framework was used as an interpretive evaluation lens to examine the potential progression from Attention and Interest to Search, Action, and Share, particularly in relation to the capacity of visual product content and promotional materials to stimulate digital engagement and tourism-related responses (Abdurrahim et al., 2019). The methodological distinctiveness of the study lies in integrating physical waste transformation, iterative visual-design development, simulated market validation, and AISAS-oriented communication analysis within a single empirical development cycle, allowing the proposed Eco-Visual Branding Loop to be grounded in observable production and market evidence rather than formulated solely as a conceptual proposition.

RESULTS AND DISCUSSION

Problem and Needs Identification (Discover Stage)

The Discover stage identified plastic bottle caps as one of the most accessible forms of waste around Lake Perintis and as a material with potential for small-scale creative transformation. Direct observation showed that discarded bottle caps were relatively easy to collect, separate, and prepare compared with larger or more contaminated plastic waste. Informal discussions with destination managers indicated that plastic waste was commonly gathered and, in some cases, burned as a practical way to reduce visible accumulation. Although burning can temporarily improve the visual cleanliness of a destination, it eliminates the possibility of recovering material value and creates additional environmental concerns. Suryawan et al. (2024) note that waste management in Indonesian tourist attractions continues to face practical and policy-related challenges, particularly in relation to plastic waste. The condition at Lake Perintis therefore presented not only a waste-management problem but also an opportunity to reconsider discarded plastic as a resource for destination-related creative production.

The material assessment further indicated that bottle caps were suitable for a micro-scale upcycling process because of their relatively compact form, manageable sorting requirements, and potential to produce varied visual effects after thermal transformation. The material needed to be sufficiently clean, workable, and durable to support repeated experimentation without creating excessive production complexity. These requirements were important because the project was designed around student-based production rather than industrial manufacturing with specialised equipment. The circular economy perspective positions waste recovery as a means of retaining material value while reducing dependence on newly extracted resources (Rodríguez et al., 2020). Such an approach shifts the role of waste from an endpoint of consumption toward an input for new forms of economic and cultural value. For Lake Perintis, the transformation of bottle caps could connect environmental management with creative production without requiring a large-scale technological system. The feasibility of this material choice consequently became an important foundation for developing a souvenir concept that could be replicated through relatively accessible production practices.

The Discover stage also revealed a gap in the tangible representation of Lake Perintis within the visitor experience. The destination provides opportunities for scenic appreciation, photography, and recreational activities, yet these experiences do not automatically produce a physical object that visitors can retain after leaving the site. Souvenirs can function as material representations of locality because they allow visitors to carry aspects of a destination into their post-trip environment (Liu et al., 2025). The absence of a distinctive product therefore represented a branding issue as well as an opportunity

for extending destination experiences beyond the physical location. A souvenir derived from local waste could address both dimensions by linking the identity of the destination with an environmental narrative. The product would not merely reproduce a visual symbol of Lake Perintis but would also communicate a transformation process associated with the place. This combination created a basis for developing an eco-visual branding approach in which material origin, visual identity, and visitor experience were considered together.

The target audience analysis then focused on users who were likely to value products that combine visual appeal, practical function, and accessible pricing. University students, young adults, and domestic visitors were considered relevant audiences because compact accessories can be incorporated into everyday activities without requiring substantial financial commitment. The product therefore needed to be lightweight, portable, visually distinctive, and sufficiently versatile to function as an accessory for keys, bags, or other personal items. A keychain was selected because its small size could reduce material consumption while allowing multiple design iterations during the prototype stage. Its everyday utility also created opportunities for the destination identity to remain visible after the visitor had left Lake Perintis. Norman (2004) argues that the emotional dimension of everyday objects can influence how users perceive and interact with designed products, making functional objects relevant carriers of symbolic and experiential meaning. The selection of a keychain consequently connected production feasibility with the need for an object capable of carrying both aesthetic and place-related meanings.

The identified problems and needs established three interconnected design requirements: environmental relevance, destination distinctiveness, and consumer usability. Environmental relevance required the product to demonstrate a credible transformation of discarded plastic rather than simply using sustainability as a promotional claim. Destination distinctiveness required the visual language to establish a recognisable relationship with Lake Perintis without reducing local identity to decorative imagery. Consumer usability required the resulting object to be practical enough to encourage actual use rather than function only as a symbolic environmental artefact. Brown (2008) describes design thinking as an iterative process in which user needs and contextual problems are translated into opportunities for experimentation and solution development. Applied to Lake Perintis, this orientation meant that waste availability alone was insufficient to determine the product concept. The material, audience, place identity, and intended use had to be considered as a connected design problem before visual identity development could begin.

The Discover findings ultimately positioned plastic waste as both an environmental liability and a potential branding resource for Lake Perintis. The most relevant opportunity was not simply to produce a souvenir from recycled material, but to develop an object in which material transformation itself contributed to the meaning of the destination. Such an approach could strengthen the relationship between ecological practice and destination representation while maintaining attention to affordability and everyday utility. Kusumaningrum (2024) shows that sustainability can contribute to destination marketing value when it is connected to tourist perceptions and experiences rather than communicated as an isolated environmental message. The Lake Perintis context suggests that this connection can be pursued through a product that combines local visual references, practical function, and visible material transformation. The findings from the Discover stage therefore provided the conceptual and practical basis for defining the identity of *Perintis by Aruna* and determining how its visual language would communicate both locality and ecological value.

Formulating the Identity of *Perintis by Aruna* (Define Stage)

The Define stage translated the needs identified during the Discover stage into a specific identity for the proposed souvenir product. The main design direction positioned the product as an eco-visual representation of Lake Perintis rather than as a generic recycled craft. The brand name Perintis by Aruna was selected to connect the destination name with a symbolic idea of renewal and transformation. *Aruna*, derived from Sanskrit and associated with dawn or reddish light, represents the emergence of new value from material previously regarded as waste. Brown (2008) explains that design thinking involves reframing contextual problems into opportunities for developing solutions that respond to user and environmental conditions. The naming process therefore established a conceptual connection between the destination, material transformation, and the intended value of the product.

The tagline “From Waste to Worth” was formulated as the central message supporting the identity of *Perintis by Aruna*. The phrase directly communicates the conversion of discarded plastic bottle caps into products with functional, aesthetic, and economic value. Its concise wording also allows the message to be consistently incorporated into product presentation and promotional materials. Sustainability communication in tourism can become more meaningful when environmental concerns are incorporated into memorable experiences rather than communicated only as abstract environmental claims (Hanna et al., 2018). The product was consequently designed so that the sustainability message would operate together with its visual appearance and practical function. Kusumaningrum (2024) shows that sustainability can contribute to destination marketing value when it is connected with tourist perceptions and experiences. The tagline thus functioned as a concise narrative device that connected the physical transformation of plastic with the broader experience offered by the destination.

The visual identity was developed through selected representations of the environmental characteristics of Lake Perintis. The design incorporated the lake landscape, an iconic ship structure, and aquatic biota such as fish, octopus, and seahorse forms. These elements were simplified into a clean contemporary cartoon aesthetic to maintain recognizability while accommodating the relatively small dimensions of the keychain. Prawira et al. (2023) indicate that visual branding can shape tourists’ responses to destination representations when visual elements create meaningful associations with place. The selected motifs were therefore treated as visual interpretations of the destination rather than literal reproductions of its physical environment. Freire et al. (2022) emphasise the need for careful consideration of cultural representation in destination branding. The visual system maintained this consideration by using local environmental references as design inspiration while avoiding claims that the selected motifs represented the entirety of local cultural identity.

The relationship between the selected motifs and the physical product can be observed in the prototype variations presented in Figure 2. The different forms demonstrate how lake-related imagery was translated into compact objects while retaining the distinctive characteristics of the recycled plastic material. The organic colour patterns generated through the transformation process also contributed variation between individual pieces, preventing the prototypes from appearing completely uniform. Such variation can strengthen the perception that each product contains a material history rather than functioning as an industrially identical commodity. Norman (2004) associates everyday product experiences with emotional responses generated through form, appearance, and interaction. In this context, the visual character of the keychain supported both functional usability and emotional associations with the destination. The prototype forms therefore became an important physical expression of the *Perintis by Aruna* identity.



Figure 2. Keychain Prototype Variations Made from Plastic Bottle Caps

Source: Personal documentation (2025)

The identity was further extended through packaging so that the environmental narrative remained connected to the product after fabrication. Eco-friendly paper cardstock was designed with lake-related imagery and the statement “Cintai Bumi, Dukung UMKM Lokal” (“Love the Earth, Support Local Enterprises”). The reverse side incorporated the narrative “Dari sampah jadi kisah” (“From waste to story”), explicitly connecting the discarded material with a new destination-related

story. Figure 3 shows the products after the finishing and packaging process, illustrating the integration of the physical product with its supporting visual communication. Destination brand authenticity can be strengthened when online and offline representations provide experiences that are perceived as meaningfully connected to the destination (Jiménez-Barreto et al., 2020). Packaging consequently functioned not only as a protective component but also as a storytelling medium that explained the product's environmental and local relevance.



Figure 3. Products Following the Finishing and Packaging Process

Source: Personal documentation (2025)

The Define stage established *Perintis by Aruna* as an integrated identity comprising the brand name, tagline, visual motifs, colour palette, material characteristics, packaging, and ecological narrative. These elements were deliberately connected so that the product could communicate locality through both its appearance and the origin of its material. The approach differs from merchandise concepts that rely only on destination images because the recycled bottle cap itself becomes part of the narrative being communicated. Wang et al. (2025) conceptualise destination branding as an increasingly interconnected process involving multiple elements and forms of participation. The Lake Perintis concept similarly connects environmental material, visual representation, product utility, and storytelling within a single identity system. The defined identity provided the design parameters for the next stage, in which the selected concepts were further developed through material experimentation, moulding, finishing, and prototype refinement.

Material Transformation and Prototype Development (Develop Stage)

The Develop stage translated the visual and conceptual identity established during the Define stage into physical products through direct experimentation with discarded plastic bottle caps. Material collection was conducted from 19 to 28 October 2025, followed by the fabrication process from 15 November to 16 December 2025. The collected bottle caps were first sorted according to colour and material characteristics to reduce inconsistencies during subsequent processing. Each selected piece was then washed and dried thoroughly to remove dirt and residual substances that could interfere with melting and moulding. The preparation process was particularly important because the quality of recycled material directly influenced the appearance and workability of the resulting prototypes. Wahida and Fuad (2022) demonstrate that plastic waste can be transformed into souvenir products when material preparation is integrated with appropriate creative production techniques. At Lake Perintis, this initial preparation established the material condition required to move from waste collection toward controlled product experimentation.

The transformation process involved heat melting followed by manual compression and moulding to produce the desired keychain forms. After the softened plastic reached a workable condition, it was compressed into moulds and allowed to form according to the predetermined product dimensions and visual concepts. The resulting pieces were subsequently trimmed along their edges to remove excess material and improve their physical finish. Hardware components were then attached so that the transformed plastic could function as an actual keychain rather than remaining a decorative plastic sample. Surface finishing was conducted to improve the overall appearance while retaining visible characteristics generated during the recycling process. Syaputra et al. (2024) demonstrate the

potential of waste-based materials to support sustainable souvenir development when material transformation is connected with tourism branding objectives. The process at Lake Perintis extended this principle by combining recycled material processing with a destination-specific visual identity developed in the preceding stage.

One notable result of the experimentation was the emergence of organic colour swirls and irregular visual patterns within individual pieces. These patterns were not treated as defects requiring complete standardisation because their variation reflected the physical characteristics of the recycled bottle-cap material. Each prototype consequently possessed a slightly different visual composition even when similar mould forms were used. Such variation provided an additional aesthetic characteristic that distinguished the products from conventionally manufactured souvenirs with highly uniform surfaces. Norman (2004) argues that the appearance and sensory character of everyday objects can contribute to users' emotional responses and perceptions of value. In this context, material irregularity became part of the product narrative because it visually signalled that the object originated from transformed waste. The design process therefore demonstrated that controlled imperfection could support the authenticity of an upcycled product while remaining compatible with the intended visual identity.

The prototyping process also exposed several technical and operational constraints that influenced the feasibility of small-scale production. Manual thermal management required careful handling because differences in heating conditions could affect the plastic's consistency, surface quality, and final form. Mould capacity limited the number of pieces that could be processed simultaneously, while sorting and preparation increased the labour required for each individual product. These conditions indicate that the prototype process cannot yet be interpreted as equivalent to standardised industrial manufacturing. The circular economy literature emphasises that waste valorisation in tourism requires attention not only to environmental benefits but also to the practical organisation of production and resource use (Rodríguez et al., 2020). The current production process therefore functioned primarily as an experimental demonstration of material feasibility and design adaptability. Its limitations simultaneously identified the technical aspects that would require further standardisation if the product were developed beyond the prototype scale.

The development stage also demonstrated that material transformation could function as part of the communication value of the product rather than merely as a production technique. The recycled bottle caps retained visible traces of their previous material form while being reorganised into objects associated with Lake Perintis through colour, shape, and visual motifs. This transformation created a direct relationship between the environmental problem identified in the Discover stage and the destination identity formulated in the Define stage. The resulting keychain therefore carried two interconnected meanings: it represented a physical response to plastic waste and served as a portable representation of the destination. Jones and Wynn (2022) emphasise the relevance of circular economy principles to tourism through approaches that retain resource value while connecting environmental management with tourism activities. The Lake Perintis prototypes operationalised this relationship at a micro scale by converting locally collected waste into a product that could circulate within the visitor economy. Material origin consequently became an active component of product meaning rather than remaining an invisible background to the branding process.

From a development perspective, the prototypes provided sufficient evidence that discarded bottle caps could be transformed into functional and aesthetically distinctive souvenirs within an accessible small-scale production setting. The process also showed that product development required continuous adjustment between material behaviour, mould dimensions, visual composition, finishing quality, and intended use. Rather than treating the first production attempt as a final product specification, the prototypes served as physical iterations through which design assumptions could be tested and refined. This iterative character corresponds with the experimental orientation of design thinking, where tangible prototypes help reveal practical problems that cannot always be identified during conceptual planning (Brown, 2008). At the same time, the absence of formal thermal safety assessment, tensile-strength testing, and long-term durability testing means that the products should remain classified as functional design prototypes rather than fully validated commercial goods. Future development would benefit from documented temperature parameters, personal protective equipment procedures, quality-control criteria, and systematic management of residual material to strengthen production safety and measurable zero-waste performance. The completed prototypes then provided the

material basis for the Deliver stage, where their visual, functional, and communicative value was introduced to a simulated market environment.

Market Testing, Dissemination, and Eco-Visual Branding Loop (Deliver Stage)

The Deliver stage introduced the completed prototypes into a controlled simulated market environment through a campus exhibition involving university students, academic staff, and visitors as a purposively selected audience. The exhibition provided an opportunity to observe how the product was received when presented together with other tourism and cultural visual communication outputs. Student creators participated directly in dissemination by providing rotation-based oral explanations of the material transformation process, visual concept, ecological rationale, and relationship between the product and Lake Perintis. This arrangement allowed the prototype to be evaluated not only as a physical souvenir but also as a communication object whose meaning depended partly on how its production story was conveyed. The exhibition setting also created a relatively accessible environment for observing audience attention, inquiries, purchasing interactions, and spontaneous responses without claiming to represent the wider tourist market. Such an approach is consistent with the role of destination-oriented social media and marketing communication in connecting visual representations with audience responses and subsequent actions (Abdurrahim et al., 2019). The Deliver stage therefore functioned as an initial market and communication test rather than as a comprehensive assessment of commercial demand.

The observed interactions could be interpreted through the AISAS sequence of Attention, Interest, Search, Action, and Share. Attention was initially generated by the vibrant colour combinations, distinctive organic patterns, and compact forms of the keychains, while Interest developed as visitors examined the products and asked about their unusual material composition. Search-related behaviour appeared through questions concerning how bottle caps were processed, why the material was selected, and how the products related to the environmental condition of Lake Perintis. Action was observable through direct purchasing interactions, whereas Share emerged when buyers documented their purchases through photographs or short videos and connected them with the project or destination through social media. Sasmita and Achmadi (2022) identify AISAS as a relevant framework for understanding marketing communication in digital environments where consumer behaviour can extend from attention and information seeking toward sharing. Fadlurrahman et al. (2023) similarly demonstrate the relevance of AISAS for tourism-oriented digital branding, particularly when visual content and visitor engagement are connected. The observed sequence suggests that the physical prototype could initiate a communication process that extended beyond the immediate exhibition encounter.

Table 2. Summary of Prototype Production and Sales Performance

Total Inventory Produced	12 Units	Micro-Scale Prototype Production Run
Unit Selling Price	Rp20,000 per unit	Accessible impulse-buy positioning
Units Sold	12 unit (100% sell-out)	Complete market absorption in simulated testing
Gross Revenue	Rp240,000	Initial economic value generation demonstrated
Testing Environment	Campus Exhibition (Simulated Market)	Controlled evaluation with purposive target audience

Source: Research Data (2025)

The sales results provide a concrete indication of initial product acceptance within the limited testing environment. As presented in Table 1, twelve keychains were produced at a selling price of IDR 20,000 per unit, and all twelve units were sold, generating gross revenue of IDR 240,000 during the simulated market activity. The complete sell-out indicates that the product achieved immediate purchase conversion among the purposively targeted exhibition audience at the tested price point. This result is relevant because the purchase decision occurred in an environment where ecological meaning, visual appeal, oral explanation, and physical product experience were simultaneously available.

Sustainability communication does not necessarily generate consumer response simply through environmental claims, because perceived experiential value can influence how sustainability is interpreted by tourists and consumers (Hanna et al., 2018). Tölkes (2018) likewise highlights the importance of communicating sustainability in ways that are meaningful and relevant to tourism audiences. The observed sales response therefore suggests an initial alignment between the product's functional utility, visual attractiveness, and ecological narrative, while the small production volume prevents the result from being generalised as evidence of broader market viability.

The market response also provided the empirical basis for conceptualising the Eco-Visual Branding Loop developed from the interaction between material transformation, destination identity, digital storytelling, and audience response. The process begins with ecological input in the form of plastic bottle caps collected from the destination environment, followed by material transformation into functional products. Local visual identity is then embedded through lake-related motifs, ship imagery, aquatic forms, and the selected colour palette, while digital storytelling gives the transformed material a communicative narrative through “From Waste to Worth” and “Dari sampah jadi kisah.” The resulting product enters an AISAS-oriented communication process in which visual exposure can generate interest, information seeking, purchase, and subsequent sharing. Market feedback then returns to the design and production process by revealing how audiences respond to product form, material characteristics, narrative, and price. The framework therefore conceptualises branding as a cyclical process rather than a one-directional promotional activity.

Table 3. Components of the Eco-Visual Branding Loop

Ecological Input	Converts Environmental Liability Into Material Asset	Plastic Bottle Caps Gathered From Danau Perintis
Material transformation	Alters form and purpose to establish aesthetic utility	Manual heat compression, moulding, and finishing
Local visual identity	Differentiates product through regional symbols	Lake biota icons, ship motifs, local colour palette
Digital storytelling	Contextualises material transformation with meaning	Tagline “From Waste to Worth” & packaging story “Dari sampah jadi kisah”
AISAS activation	Drives awareness, search, acquisition, and sharing	Campus exhibition, forum pitch, sales, post-purchase posts
Market feedback	Informs continuous design and process refinement	Complete sell-out data and visitor inquiries

Source: Research Data (2025)

Table 3 clarifies that each component contributes a distinct function while remaining connected to the next stage of the branding process. Ecological input establishes the material origin, while material transformation converts that origin into an object that can carry visual and functional value. Local visual identity differentiates the product from generic recycled merchandise, and digital storytelling provides the interpretive context through which consumers can understand the relationship between waste, place, and product. AISAS activation then connects the physical product with audience behaviour, while market feedback creates an opportunity for subsequent adjustments to design, production, packaging, and communication. This structure extends conventional destination branding by making the material history of the souvenir part of the brand communication itself. Jones and Wynn (2022) emphasise the relevance of circular economy principles in tourism through the retention and re-use of material value within tourism-related systems. Rodríguez et al. (2020) similarly identify circular practices as an important pathway for connecting tourism activities with resource efficiency, although the present framework remains a conceptual model derived from a micro-scale empirical process rather than a tested causal model.

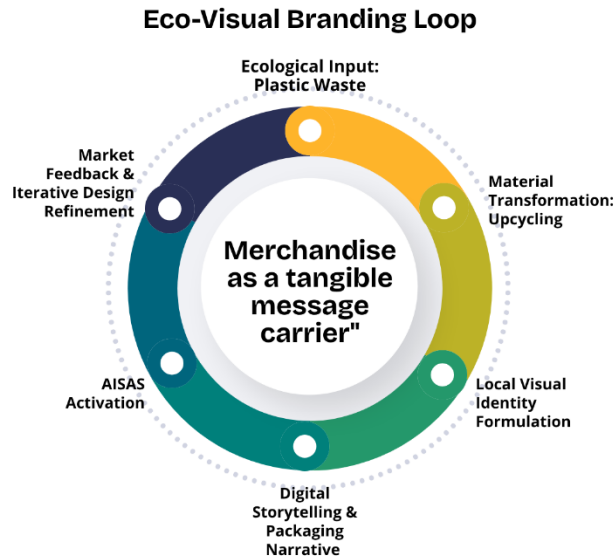


Figure 4. Structure of the Eco-Visual Branding Loop
Source: Authors' conceptual development (2026)

Figure 4 illustrates the cyclical relationship among the six components identified through the development and delivery process. The distinctive feature of the model lies in the position of transformed waste as an active communication resource rather than simply as a production input. A physical souvenir can continue to function as a reminder of the destination after the visitor leaves because it remains embedded in everyday use, while subsequent photographs or videos can convert that physical object into digital destination-related content. Alalwan et al. (2017) explain that social media marketing enables consumer interaction and content circulation that can extend communication beyond the original point of contact. In the Lake Perintis context, this relationship creates a potential bridge between circular materiality, destination branding, and digital word-of-mouth. However, the available data do not include reach, impressions, engagement rates, destination search volume, quantified user-generated content, or changes in visitation intention, so Figure 4 should be interpreted as a conceptual explanation of the observed process rather than evidence of causal increases in destination awareness. The value of the model lies in identifying a potentially iterative mechanism through which environmental material, visual identity, and consumer communication can be developed together.



Figure 5. Documentation of Product Dissemination and Exhibition by the Student Team
Source: Personal documentation (2025)

Figure 5 documents the dissemination context in which the prototypes were introduced and explained to the target audience. The exhibition demonstrates how student involvement extended beyond fabrication into product presentation, storytelling, and direct audience interaction, making

Research-Based Learning part of the delivery process. The practical implication is that destination managers and creative micro-enterprises could adapt a similar process by connecting community-based waste collection with small-scale production, visual branding, and experiential product presentation. Packaging could subsequently incorporate QR codes directing users toward artisan stories, material transformation information, environmental indicators, or responsible visitor practices, thereby extending the communication pathway beyond the point of purchase. Community-oriented plastic management can strengthen the relationship between environmental action and sustainable tourism when local participation is integrated into waste collection and destination management processes (Nizar et al., 2026). The present findings nevertheless remain bounded by the production of twelve units, a short testing period, the absence of buyer demographic profiling, and the lack of formal material stress and durability testing. Future development could examine larger-scale community co-creation, standardised production procedures, pre- and post-exposure measures of destination brand awareness, and longitudinal digital analytics to determine whether the Eco-Visual Branding Loop operates consistently beyond the simulated market context.

CONCLUSION

The study developed an upcycled tourism merchandise model from plastic bottle caps collected in the Lake Perintis context and transformed the material into functional keychains under the *Perintis by Aruna* brand. The development process demonstrated that discarded plastic could be converted into aesthetically distinctive products that combine local visual references, ecological meaning, practical utility, and potential economic value. The Double Diamond process connected waste identification, destination-oriented identity formulation, material experimentation, prototype development, and simulated market testing within an integrated design framework. Under the tagline “From Waste to Worth,” twelve prototype units were offered at IDR 20,000 per unit during a campus-based exhibition, and all twelve units were sold, generating IDR 240,000 in gross revenue. The complete sell-out provides an initial indication of product acceptance within the specific simulated market environment and demonstrates that the transformed material could attract purchase interest when combined with functional and visual value. However, the result should be interpreted as an initial proof-of-concept because the production volume was limited, the testing period was short, and the audience did not represent a statistically defined tourism market. The findings indicate that plastic waste at a tourism destination can be repositioned from an environmental burden into a tangible component of destination-related creative production.

The principal conceptual contribution is the Eco-Visual Branding Loop, which integrates ecological waste inputs, material transformation, local visual identity, digital storytelling, AISAS-oriented consumer journeys, and iterative market feedback into a cyclical branding process. The model positions upcycled merchandise not simply as a recycled product but as a tangible communication medium through which environmental action and destination identity can be experienced simultaneously. Material origin becomes part of the brand narrative because the physical object retains a visible connection with the waste problem from which it was created. The product can also extend destination communication beyond the exhibition context when consumers use, photograph, or share the souvenir through digital platforms. Future development should expand production capacity while maintaining material quality, integrate community co-creation, and establish clearer procedures for thermal safety, durability, and quality control. Further research should employ buyer profiling, formal material testing, longitudinal destination-brand awareness measures, and social media conversion analytics to examine the sustainability and communication performance of the proposed model over time. These developments would allow the Eco-Visual Branding Loop to be evaluated beyond its current micro-scale prototype context and potentially refined as a replicable approach to circular tourism branding.

REFERENCES

- Abdurrahim, M. S., Najib, M., & Djohar, S. (2019). Development of AISAS model to see the effect of tourism destination in social media. *Jurnal Aplikasi Manajemen*, 17(1), 133–143. <https://doi.org/10.21776/ub.jam.2019.017.01.15>

- Alalwan, A. A., Rana, N. P., Dwivedi, Y. K., & Algharabat, R. (2017). Social media in marketing: A review and analysis of the existing literature. *Telematics and Informatics*, 34(7), 1177–1190. <https://doi.org/10.1016/j.tele.2017.05.008>
- Brown, T. (2008). Design thinking. *Harvard Business Review*, 86(6), 84–92.
- Bruin, R., Cooper, E., Lamers, M., & Blair, B. (2025). Sustaining the brand and branding sustainability: Landscape as homeland in Greenland's visual marketing. *Journal of Sustainable Tourism*, 33(11), 2484–2503. <https://doi.org/10.1080/09669582.2024.2400572>
- Chen, R., Zhou, Z., Zhan, G., & Zhou, N. (2020). The impact of destination brand authenticity and destination brand self-congruence on tourist loyalty: The mediating role of destination brand engagement. *Journal of Destination Marketing & Management*, 15, 100402. <https://doi.org/10.1016/j.jdmm.2019.100402>
- Fadlurrahman, M. A., Rochmah, L. M., & Pahlevi, A. S. (2023). Video branding as AISAS mode's digital marketing strategy for Hijau Daun mangrove ecotourism in Bawean Island. *Publica: Jurnal Pemikiran Administrasi Negara*, 15(2), 485–497. <https://doi.org/10.15575/jpan.v15i2.28866>
- Freire, J., Gertner, R., & Gertner, D. (2022). Cultural appropriation and destination brands. *GeoJournal of Tourism and Geosites*, 40(1), 79–88. <https://doi.org/10.30892/gtg.40109-805>
- Hanna, P., Font, X., Scarles, C., Weeden, C., & Harrison, C. (2018). Tourist destination marketing: From sustainability myopia to memorable experiences. *Journal of Destination Marketing & Management*, 9, 36–43. <https://doi.org/10.1016/j.jdmm.2017.10.002>
- Iswahyudi, I., Sukma, D. S. R. A., Zhulva Nabila Wahyudi, Z., et al. (2026). Linking lake debris pollution, tourist acceptance, and sustainability management in a freshwater tourism destination: Evidence from Lake Koto Kampar, Indonesia. *Total Environment Advances*, 19, 200164. <https://doi.org/10.1016/j.teadva.2026.200164>
- Jiménez-Barreto, J., Rubio, N., & Campo, S. (2020). Destination brand authenticity: What an experiential simulacrum! A multigroup analysis of its antecedents and outcomes through official online platforms. *Tourism Management*, 77, 104022. <https://doi.org/10.1016/j.tourman.2019.104022>
- Jog, D., & Gumparthi, V. P. (2025). The impact of co-creation of experience and souvenir authenticity on destination brand experience and souvenir repurchase intention. *Vision*. Advance online publication. <https://doi.org/10.1177/09722629251384106>
- Jones, P., & Wynn, M. (2022). The circular economy and natural capital in the tourism and hospitality industry. *Journal of Circular Economy*, 1(1). <https://doi.org/10.55845/IRVU6231>
- Khaerunnisa, K., & Cininta, M. (2023). Program upcycling sampah plastik di kawasan Kompleks Candi Borobudur. *Jurnal Atma Inovasia*, 3(6). <https://doi.org/10.24002/jai.v3i6.7809>
- Kusumaningrum, S. D. (2024). How can sustainability add value in destination marketing? The tourist perspective. *Cogent Business & Management*, 11(1), 2425718. <https://doi.org/10.1080/23311975.2024.2425718>
- Liu, Y., Chen, Y., Font, X., Zhang, Y., & Hu, J. (2025). Echoes of a destination: The representation of locality in souvenirs and their role in sustaining post-trip, tourist well-being. *Tourism Management*, 108, 105125. <https://doi.org/10.1016/j.tourman.2024.105125>
- Nizar, M., Erdiwansyah, E., Bahagia, B., Irhamni, I., Nofaris, S., & Afifah, S. (2026). Toward a community-based plastic waste management framework to enhance sustainable tourism in small island destinations: a case study of Sabang Island, Indonesia. *Marine Development*, 4(1), 12. <https://doi.org/10.1007/s44312-026-00081-w>
- Norman, D. A. (2004). *Emotional design: Why we love (or hate) everyday things*. Basic Books.
- Prawira, N. G., Susanto, E., & Prawira, M. F. A. (2023). Visual branding on Indonesian tourism destinations: Does it affect tourists? *ABAC Journal*, 43(1). <https://doi.org/10.14456/abacj.2023.4>
- Rodríguez, C., Florido, C., & Jacob, M. (2020). Circular economy contributions to the tourism sector: A critical literature review. *Sustainability*, 12(11), 4338. <https://doi.org/10.3390/su12114338>
- Sasmita, A. S., & Achmadi, N. S. (2022). The popularity of TikTok and the implementation of the AISAS model on marketing communications through TikTok. *Manajemen Bisnis*, 12(1), 62–76. <https://doi.org/10.22219/mb.v12i01.17863>

- Suryawan, I. W. K., Suhardono, S., Sari, M. M., & Septiariva, I. Y. (2024). Waste recycling processing in Indonesian tourist attractions: Problems and challenges of implementing legal policies restricting plastic use. *Jurnal Ilmu Lingkungan*, 22(4), 1067–1077. <https://doi.org/10.14710/jil.22.4.1067-1077>
- Syaputra, E. A., Parmita, A. W. Y. P., Febrianti, N., Maharany, G. T., & Rossalina, N. (2024). Implementation of green innovation in designing sustainable tourism souvenirs using biocomposite material from waste of empty palm oil bunches (EFB) to support IKN branding as the greenest country capital in the world. *SPECTA Journal of Technology*, 8(3), 223–232. <https://doi.org/10.35718/specta.v8i3.1222>
- Syaputra, E. A., Parmita, A. W. Y. P., Febrianti, N., Maharany, G. T., & Rossalina, N. (2024). Implementation of green innovation in designing sustainable tourism souvenirs using biocomposite material from waste of empty palm oil bunches (EFB) to support IKN branding as the greenest country capital in the world. *SPECTA Journal of Technology*, 8(3), 223–232. <https://doi.org/10.35718/specta.v8i3.1222>
- Tölkes, C. (2018). Sustainability communication in tourism—A literature review. *Tourism Management Perspectives*, 27, 10–21. <https://doi.org/10.1016/j.tmp.2018.04.002>
- United Nations Environment Programme. (2023). *Turning off the tap: How the world can end plastic pollution and create a circular economy*. <https://doi.org/10.59117/20.500.11822/42277>
- Wahida, A., & Fuad, F. R. (2022). Recycling plastic waste for the development of souvenir products special for Surakarta City. *Mudra Jurnal Seni Budaya*, 37(3), 271–280. <https://doi.org/10.31091/mudra.v37i3.1805>
- Wang, Y., et al. (2025). Destination brand gestalt: Dimensionalizing co-created tourism destination branding. *International Journal of Tourism Cities*, 11(3–4). <https://doi.org/10.1108/IJTC-02-2024-0049>